

1. Product Technical Requirements

Functional Specifications (Tech-Focused)

- Overview: TODO — brief feature list derived from codebase and product backlog.
- For each feature:
 - Purpose: TODO
 - Inputs: HTTP params, events, file formats
 - Outputs: responses, events, side effects
 - Edge cases: validation, retries, failure modes
 - Sequence diagram: reference [diagrams/sequence_diagrams.mmd](#)

Example feature template

- Name: "Create Booking"
- Endpoint: `POST /api/v1/bookings`
- Inputs: `user_id (uuid)`, `slot_id (uuid)`, `metadata (object)`
- Success: 201 Created + booking object
- Errors: 400 (validation), 401 (auth), 409 (conflict)

Non-Functional Requirements

- Performance: P95 latency $\leq$ TODO ms for core endpoints
- Scalability: horizontal autoscaling for API and workers
- Security: TLS everywhere, role-based access
- Compliance: GDPR data residency & right-to-delete
- Reliability: target uptime 99.9% (SLA)

Acceptance Criteria

- Each feature must have unit + integration tests covering happy + error flows.
- API contract published in `openapi.yaml`.